

Not Recently Used Page Replacement Algorithm

1 Core Idea

Definition

The **Not Recently Used (NRU)** page replacement algorithm divides pages into classes using the **Referenced** bit R and the **Modified** bit M , then evicts a page from the lowest-numbered nonempty class.

NRU is based on a practical idea: a page that has not been referenced recently is usually a better victim than a page that is currently active.

NRU Rule

choose a random page from the lowest-numbered nonempty class

2 The R and M Bits

Most virtual-memory systems keep two usage bits in each page table entry.

Bit	Name	Meaning
R	Referenced	Set to 1 whenever the page is read or written.
M	Modified	Set to 1 whenever the page is written to.

Hardware Support

The R and M bits must be updated on memory references, so hardware usually sets them automatically.

Once a bit becomes 1, it stays 1 until the operating system clears it.

3 Why the OS Clears R

When a process starts, the OS initializes both bits:

$$R = 0, \quad M = 0$$

Then, periodically, usually on a clock interrupt, the OS clears the R bit.

Purpose of Clearing R

Clearing R lets the OS distinguish pages referenced recently from pages not referenced since the last clock tick.

The M bit is not cleared in the same way because it tells the OS whether the page must be written back to disk before eviction.

4 NRU Classes

When a page fault occurs, NRU inspects pages and sorts them into four classes.

Class	R	M	Meaning
0	0	0	Not referenced recently and not modified. Best victim class.
1	0	1	Not referenced recently but modified.
2	1	0	Referenced recently but not modified.
3	1	1	Referenced recently and modified. Worst victim class.

Class Priority

NRU prefers lower class numbers:

Class 0 $\rightarrow$ Class 1 $\rightarrow$ Class 2 $\rightarrow$ Class 3

5 Why Class 1 Can Exist

At first, class 1 may look impossible:

$$R = 0, \quad M = 1$$

If a page was modified, it must also have been referenced. So how can it be modified but not referenced?

Explanation

A page can first be in class 3, meaning $R = 1, M = 1$. Later, a clock interrupt clears R , but leaves M unchanged. The page becomes class 1: $R = 0, M = 1$.

Key Detail

Clock interrupts clear R , not M . The modified bit must remain available so the OS knows whether disk write-back is needed.

6 Algorithm Steps

1. A page fault occurs.
2. The OS checks the R and M bits of resident pages.
3. Pages are grouped into classes 0, 1, 2, and 3.
4. The OS finds the lowest-numbered class that is not empty.
5. It chooses a random page from that class.

6. If the chosen page is dirty, it is written back to disk.
7. The incoming page is loaded into the freed frame.

7 Worked Example

Problem

Suppose the pages in memory have these bit values:

Page	R	M
A	1	1
B	0	1
C	1	0
D	0	0

Which page class should NRU choose from?

Solution

Page	R	M	Class
A	1	1	Class 3.
B	0	1	Class 1.
C	1	0	Class 2.
D	0	0	Class 0.

The lowest nonempty class is class 0.

Answer

NRU chooses from class 0. In this example, it evicts page D .

8 Why NRU Prefers Class 1 over Class 2

NRU chooses class 1 before class 2, even though class 1 pages are dirty.

Reason

NRU considers “not recently referenced” more important than “clean.” A dirty page not used recently may be a better victim than a clean page that is currently active.

This is a heuristic. It is not always perfect, but it is simple and often adequate.

9 If Hardware Lacks R and M Bits

The OS can simulate these bits using page faults.

Event	OS action
First reference to a page	Cause a page fault, set simulated $R = 1$, and mark the page read-only.
First write to that page	Cause another page fault, set simulated $M = 1$, and mark the page read/write.

Cost

Simulating R and M works, but it is slower than direct hardware support because it uses page faults.

10 Strengths and Weaknesses

Point	Explanation
Strength	Easy to understand and moderately efficient.
Strength	Uses simple hardware-maintained bits.
Weakness	Chooses randomly within a class, so it may not pick the best page.
Weakness	Less accurate than algorithms that track page use more precisely.

11 Final Summary

One-Box Summary

NRU uses R and M bits to classify pages.
Lowest class is preferred: 00, 01, 10, 11.
It evicts randomly from the lowest nonempty class.
It is simple and practical, but not optimal.